

actions, extending the original framework. Other examples include the work of Gnilke and Zumbrägel [15], who linked Maze et al.'s ideas to recent advances in isogeny-based cryptography. Additionally, [16] introduced a key-exchange protocol based on twisted group rings, incorporating a group key-exchange agreement.

In this research, Grigoriev and Shpilrain proposed the use of tropical semiring for public key exchange [2], [3] and for digital signatures [4]. Nevertheless, the first attempt was analysed by [5], where it was introduced the so called Kotov-Ushakov attack, an heuristic attack that nowadays has become the standar attack against tropical cryptography. In [6], the authors propose a new deterministic attack against a public key exchange protocol based on tropical semiring, that improve the Kotov-Ushakov attack in the sense that it can be used in the same scenarios, but with a deterministic output. In addition, the other two tropical cryptography protocols proposed by Grigoriev and Shpilrain has been proven not secure in [9], [7] and [8].

Far from being abandoned, this ideas has been further explored. In [17], the authors recently proposed the use of triad matrix semiring. This semiring is a generalization of tropical semiring where elements are a vector of 3 entries, and with a modified addition and multiplication that endow them with the structure of semiring. Based on them, the authors establish a public key exchange protocol that use matrix with entries over triad semiring.

2 Results

In this paper, we introduce an isomorphism between triad semiring and circulant matrix over tropical semiring. As a result, it is possible to establish an isomorphism between triad matrix semiring and tropical matrix semiring, and therefore reinterpretate the public key exchange in terms of tropical matrix. This prove that the previous protocol can be seen as the Stickel protocol where instead of taking a polynomial, only a monomial is used. Finally, this protocol is susceptible to the attack introduced in [18].

3 Preliminaries

In this section we will introduce some basic background on tropical semiring as well as triad semiring.

Definition 1 A semiring R is a non-empty set together with two operations $+$ and $\cdot$ such that $(S, +)$ is a commutative monoid, $(S, \cdot)$ is a monoid and the following distributive laws hold:

$$\begin{aligned} a(b + c) &= ab + ac \\ (a + b)c &= ac + bc \end{aligned}$$

We say that $(R, +, \cdot)$ is additively idempotent if $a + a = a$ for all $a \in R$.

Definition 2 Let R be a semiring and $(M, +)$ be a commutative semigroup with identity 0_M . M is a right semimodule over R if there is an external operation $\cdot : M \times R \rightarrow M$ such that